

step-1: Import packages

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step-2: Create a Dataframe using list

```
In [6]: Names=['Ramesh','Suresh','Sathish']
#pd.DataFrame(data,index,columns)
pd.DataFrame(Names)
```

```
Out[6]:
```

	0
0	Ramesh
1	Suresh
2	Sathish

Step-3:Name the column

```
In [ ]: Names=['Ramesh','Suresh','Sathish']
#pd.DataFrame(data,index,columns)
pd.DataFrame(Names,'NAMES') # w
pd.DataFrame(Names,columns='NAMES') # wrong
pd.DataFrame(Names,['NAMES']) # wrong
pd.DataFrame(Names,columns=['NAMES']) # right

'Names','Age'
['Names','Age']
```

```
In [8]: Names=['Ramesh','Suresh','Sathish']
#pd.DataFrame(data,index,columns)
pd.DataFrame(Names,columns=['Names'])
```

```
Out[8]:
```

	Names
0	Ramesh
1	Suresh
2	Sathish

step-4: Change the index

```
In [15]: Names=['Ramesh','Suresh','Sathish']
#pd.DataFrame(data,index,columns)
pd.DataFrame(Names,
             index=['A','B','C'],
             columns=['Names'])
```

Out[15]:

	Names
A	Ramesh
B	Suresh
C	Sathish

Step-5: two columns

```
In [18]: Names=['Ramesh','Suresh','Sathish']
Age=[20,22,24]
```

```
In [33]: data=zip(Names,Age)
cols=['Names','Age']
df=pd.DataFrame(data,columns=cols)
len(df)
```

Out[33]: 3

Step-6: Add the data to empty dataframe

```
In [31]: df=pd.DataFrame()
df
```

Out[31]: —

```
In [35]: # selection or update = []
# function = ()
data=zip(Names,Age)
cols=['Names','Age']
employee=pd.DataFrame(data,columns=cols)
```

```
In [39]: employee['Names']
```

Out[39]:

0	Ramesh
1	Suresh
2	Sathish

Name: Names, dtype: object

```
In [47]: df=pd.DataFrame()
df['Names']=['Ramesh','Suresh','Sathish']
df['Age']=[20,22,24]
```

```
In [49]: df
```

Out[49]:

	Names	Age
0	Ramesh	20
1	Suresh	22
2	Sathish	24

- key error : No column

- dim error : column length is mismatch

In [52]: df

Out[52]:

	Names	Age
0	Ramesh	20
1	Suresh	22
2	Sathish	24

step-7: add the column to existed dataframe

In [59]: df['City']=['Pune','Mumbai','Delhi']
df

Out[59]:

	Names	Age	City
0	Ramesh	20	Pune
1	Suresh	22	Mumbai
2	Sathish	24	Delhi

Step-8:Rename the column

In [62]: *# I want to rename 'City' to 'city'*
#df.rename mapper, index, columns, axis, inplace)
d={'City':'city'}
df.rename(columns=d)

Out[62]:

	Names	Age	city
0	Ramesh	20	Pune
1	Suresh	22	Mumbai
2	Sathish	24	Delhi

In [72]: df.rename(d,axis=0) *# assume as axis=0 I need to change index No answer-No error*
df.rename(d,axis=1)

Out[72]:

	Names	Age	city
0	Ramesh	20	Pune
1	Suresh	22	Mumbai
2	Sathish	24	Delhi

In [74]: d={'City':'city'} *# Column*
df.rename(d) *# index*
df.rename(d,axis=0) *# index*
df.rename(d,axis=1) *# col == work*

```
df.rename(index=d) # in
df.rename(columns=d) # column == work
```

Out[74]:

	Names	Age	city
0	Ramesh	20	Pune
1	Suresh	22	Mumbai
2	Sathish	24	Delhi

In [86]: `df.rename(d,axis=1,inplace=True)`

In [88]: `df`

Out[88]:

	Names	Age	city
0	Ramesh	20	Pune
1	Suresh	22	Mumbai
2	Sathish	24	Delhi

In [82]: `l=[20,10,30,40,50]`
`import random`
`random.shuffle(l)`

In [84]: `l`

Out[84]: `[30, 20, 40, 10, 50]`

In [92]: `df.columns=['names','age','City']`

In [96]: `df.index=['A','B','C']`

In [98]: `df`

Out[98]:

	names	age	City
A	Ramesh	20	Pune
B	Suresh	22	Mumbai
C	Sathish	24	Delhi

Step-9: Drop the column

In [103... `# I want to drop 'City column'`
`df.drop('City') # error`
`df.drop('City',axis=0) # error`
`df.drop(index=['City']) # error`
`df.drop('City',axis=1) # work`
`df.drop(columns=['City']) # work`

```

-----
KeyError                                Traceback (most recent call last)
Cell In[103], line 2
      1 # I want to drop 'City column'
----> 2 df.drop('City')

File E:\Anaconda\Lib\site-packages\pandas\core\frame.py:5581, in DataFrame.drop(self, labels, axis, index, columns, level, inplace, errors)
    5433 def drop(
    5434     self,
    5435     labels: IndexLabel | None = None,
    (... )
    5442     errors: IgnoreRaise = "raise",
    5443 ) -> DataFrame | None:
    5444     """
    5445     Drop specified labels from rows or columns.
    5446     (...)
    5579         weight  1.0      0.8
    5580     """
-> 5581     return super().drop(
    5582         labels=labels,
    5583         axis=axis,
    5584         index=index,
    5585         columns=columns,
    5586         level=level,
    5587         inplace=inplace,
    5588         errors=errors,
    5589     )

File E:\Anaconda\Lib\site-packages\pandas\core\generic.py:4788, in NDFrame.drop(self, labels, axis, index, columns, level, inplace, errors)
    4786 for axis, labels in axes.items():
    4787     if labels is not None:
-> 4788         obj = obj._drop_axis(labels, axis, level=level, errors=errors)
    4790 if inplace:
    4791     self._update_inplace(obj)

File E:\Anaconda\Lib\site-packages\pandas\core\generic.py:4830, in NDFrame._drop_axis(self, labels, axis, level, errors, only_slice)
    4828     new_axis = axis.drop(labels, level=level, errors=errors)
    4829     else:
-> 4830     new_axis = axis.drop(labels, errors=errors)
    4831     indexer = axis.get_indexer(new_axis)
    4833 # Case for non-unique axis
    4834 else:

File E:\Anaconda\Lib\site-packages\pandas\core\indexes\base.py:7070, in Index.drop(self, labels, errors)
    7068 if mask.any():
    7069     if errors != "ignore":
-> 7070         raise KeyError(f"{labels[mask].tolist()} not found in axis")
    7071     indexer = indexer[~mask]
    7072     return self.delete(indexer)

KeyError: "['City'] not found in axis"

```

In [115...

```

Names=['Ramesh','Suresh','Sathish']
City=['Hyd','Blr','Pune']
Age=[20,22,24]

```

```
data=zip(Names, City, Age)
df=pd.DataFrame(data,
                 index=['A', 'B', 'C'],
                 columns=['Names', 'City', 'Age'])
#####
df.drop('City', axis=1, inplace=True)
df.drop('C', inplace=True)
```

In [117...

df

Out[117...

	Names	Age
A	Ramesh	20
B	Suresh	22

- creating a dataframe
- Name the column
- Change the Index
- add the columns
- add the data to empty dataframe
- add the column to existing dataframe
- rename the column
- rename all the column and all the index at a time
- drop the column
- drop the index

In [19]:

```
# create a dataframe
import pandas as pd
df=pd.DataFrame()
df['Names']=['Ramesh', 'Suresh', 'Sathish']
df['Age']=[20, 22, 24]
df['City']=['Hyd', 'Blr', 'Pune']
df
```

Out[19]:

	Names	Age	City
0	Ramesh	20	Hyd
1	Suresh	22	Blr
2	Sathish	24	Pune

Step-10:replace

```
In [7]: #df.replace(to_replace,value)
df.replace('Hyd','Chennai') # inplace not happend so it will not overwrite
```

```
Out[7]:
```

	Names	Age	City
0	Ramesh	20	Chennai
1	Suresh	22	Blr
2	Sathish	24	Pune

```
In [9]: df
```

```
Out[9]:
```

	Names	Age	City
0	Ramesh	20	Hyd
1	Suresh	22	Blr
2	Sathish	24	Pune

```
In [15]: df['Names']='Mahesh'
df['Age']=24
df['City']='Hyd'
```

```
In [17]: df
```

```
Out[17]:
```

	Names	Age	City
0	Mahesh	24	Hyd
1	Mahesh	24	Hyd
2	Mahesh	24	Hyd

```
In [21]: import pandas as pd
df=pd.DataFrame()
df['Names']=['Ramesh','Suresh','Sathish','Mahesh']
df['Age']=[20,22,24,24]
df['City']=['Hyd','Blr','Pune','Hyd']
df
```

```
Out[21]:
```

	Names	Age	City
0	Ramesh	20	Hyd
1	Suresh	22	Blr
2	Sathish	24	Pune
3	Mahesh	24	Hyd

```
In [23]: df.replace('Hyd','Chennai')
```

```
Out[23]:
```

	Names	Age	City
0	Ramesh	20	Chennai
1	Suresh	22	Blr
2	Sathish	24	Pune
3	Mahesh	24	Chennai

```
In [25]: df['Marks']=[25,24,23,22]
```

```
In [27]: df
```

```
Out[27]:
```

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	24
2	Sathish	24	Pune	23
3	Mahesh	24	Hyd	22

```
In [29]: df.replace(24,20)
```

```
Out[29]:
```

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	20
2	Sathish	20	Pune	23
3	Mahesh	20	Hyd	22

```
In [31]: # all the data in same will change  
# all the column data will change  
  
df
```

```
Out[31]:
```

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	24
2	Sathish	24	Pune	23
3	Mahesh	24	Hyd	22

```
In [35]: df['Marks'].replace(24,20,inplace=True)
```

```
In [37]: df
```

Out[37]:

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	20
2	Sathish	24	Pune	23
3	Mahesh	24	Hyd	22

```
In [ ]: # I want to change city name to Chennai
# for the candidates who has age = 24
```

step11-np.where

```
In [41]: import numpy as np
```

```
In [ ]: np.where(condition,True,False)
age==24
Chennai
Nochange
```

```
In [49]: # I need to apply condition on age column
# we need to select a column
con=df['Age']==24
true='Chennai'
false=df['City']
```

```
In [57]: df['City']=np.where(con,true,false)
```

```
In [59]: df
```

Out[59]:

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	20
2	Sathish	24	Chennai	23
3	Mahesh	24	Chennai	22

```
In [ ]: # Big4
# Deloitte, EY, PWC,KPMG
# SQL PANDAS
# MAANG : Micr AMA APPL NV G
# developer
# Skills DATA Structures
# create products
# SQL vs Pandas
```

Step-12: Append a row

```
In [62]: df
```

Out[62]:

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	20
2	Sathish	24	Chennai	23
3	Mahesh	24	Chennai	22

loc-iloc

In []: `df.loc[rows,columns]`
 rows are index numbers similar to list slice
`l[start:stop:step]`
 column are names any column name we can provide
`df.iloc[rows,columns]`
 rows and columns are index numbers similar to list slice
`l[start:stop:step]`

In [73]: `rows=[1,2,3]`
`cols=['Age','City']`
`df.loc[rows,cols]`

Out[73]:

	Age	City
1	22	Blr
2	24	Chennai
3	24	Chennai

In [75]: `df`

Out[75]:

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	20
2	Sathish	24	Chennai	23
3	Mahesh	24	Chennai	22

In [77]: `rows=[1,2,3]`
`cols=[1,2]`
`df.iloc[rows,cols]`

Out[77]:

	Age	City
1	22	Blr
2	24	Chennai
3	24	Chennai

In [79]: `cols=['Age','City']`

```
df.loc[1:3,cols]
```

```
Out[79]:
```

	Age	City
1	22	Blr
2	24	Chennai
3	24	Chennai

```
In [89]: df.loc[len(df)]=['Anish',26,'Mumbai',25]
```

```
In [91]: df
```

```
Out[91]:
```

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	20
2	Sathish	24	Chennai	23
3	Mahesh	24	Chennai	22
4	Anish	26	Mumbai	25

step-13:How to save the dataframe

```
In [95]: # directory : where is the file available
# name      : what is the name of the file
# extension : what is the type of the file

#####
#      SAME PYTHON LOCATION      #
# name      : what is the name of the file
# extension : what is the type of the file

# dataframe variable = df
# name : student
# exten : .csv and .xlsx

df.to_csv('student.csv')
```

```
In [97]: pd.read_csv('student.csv')
```

```
Out[97]:
```

	Unnamed: 0	Names	Age	City	Marks
0	0	Ramesh	20	Hyd	25
1	1	Suresh	22	Blr	20
2	2	Sathish	24	Chennai	23
3	3	Mahesh	24	Chennai	22
4	4	Anish	26	Mumbai	25

Step-14:Ignore index

```
In [100... df.to_csv('student.csv',index=False)
```

```
In [102... pd.read_csv('student.csv')
```

```
Out[102...
```

	Names	Age	City	Marks
0	Ramesh	20	Hyd	25
1	Suresh	22	Blr	20
2	Sathish	24	Chennai	23
3	Mahesh	24	Chennai	22
4	Anish	26	Mumbai	25

```
In [104... df.to_excel('student.xlsx')
```

```
In [106... pd.read_excel('student.xlsx')
```

```
Out[106...
```

	Unnamed: 0	Names	Age	City	Marks
0	0	Ramesh	20	Hyd	25
1	1	Suresh	22	Blr	20
2	2	Sathish	24	Chennai	23
3	3	Mahesh	24	Chennai	22
4	4	Anish	26	Mumbai	25

```
In [ ]: # take a new notebook  
# in your datafiles  
# visadataset  
# bankdataset pls load that
```